

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools : Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Scenario-Based Assessment

Topic 1: Footprinting Tools – Conducting Competitive Intelligence

Q1. (L3 – Apply) – 15 Marks

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named ABC Technologies. The consultant must collect publicly available information without directly interacting with the company's servers.

Task: Apply suitable footprinting tools to identify the company's:

- domain registration details,
- employee email addresses,
- technology stack,
- and publicly exposed subdomains.

Explain how each selected tool helps in conducting competitive intelligence.

Tools Used :

1. WHOIS
2. theHarvester
3. Wappalyzer
4. BuiltWith
5. Crt.sh

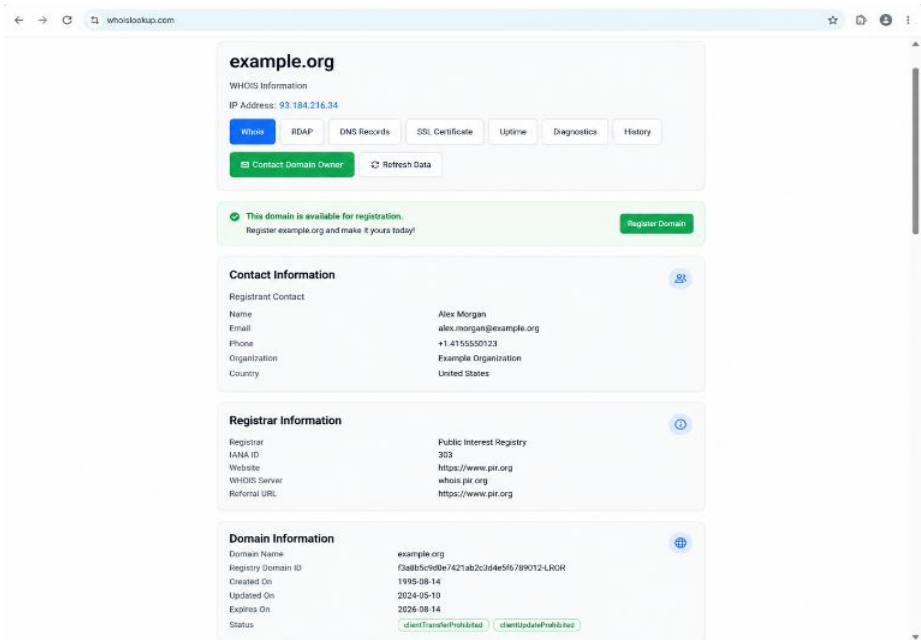
Procedure :

- Retrieved domain registration details using WHOIS.
- Enumerated publicly available employee email addresses using theHarvester.
- Identified the website technology stack using Wappalyzer and BuiltWith.
- Discovered publicly exposed subdomains using crt.sh.

Results :

1. Domain Registration Details :

- Domain: abc.com
- Registrar: CSC Corporate Domains, Inc.



WHOIS data last fetched July 11, 2028	
Nameservers	
Hostname	IP Address
ns-1368.awsdns-43.org	205.251.197.88
ns-1869.awsdns-41.co.uk	205.251.199.77
ns-318.awsdns-39.com	205.251.193.62
ns-736.awsdns-28.net	205.251.194.224
Domain Status	
clientTransferProhibited https://icann.org/epp#clientTransferProhibited	
serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited	
serverTransferProhibited https://icann.org/epp#serverTransferProhibited	
serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited	

2. Employee Email Enumeration

- Public employee email addresses were identified
- Public hosts and IP addresses were also discovered.

[illegible]

```

[+] IPs found: 31
13.226.289.25
13.226.289.79
13.226.289.93
13.226.289.93
13.226.289.180
13.226.244.67
13.226.244.53
13.226.244.75
131.153.151.35
162.288.138.132
172.66.46.28
172.66.67.60
18.168.96.126
18.168.96.10
18.168.96.26
18.168.96.78
18.66.187.9
192.0.76.12
2486196c131120::3
3.162.132.113
3.162.132.30
3.162.132.23
3.162.3.100
3.162.3.110
3.162.3.125
3.162.3.40
3.170.45.75
3.773.161.101
66.256.176.22
66.256.62.150
99.64.92.26

[+] Hosts found: 1
diana.phlomagabc.com

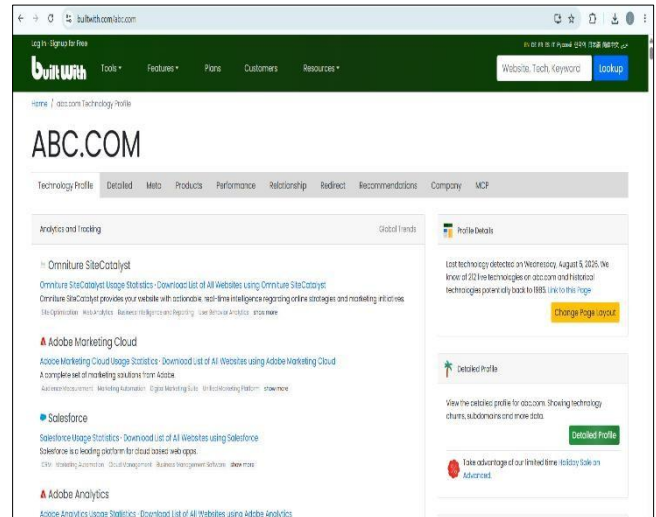
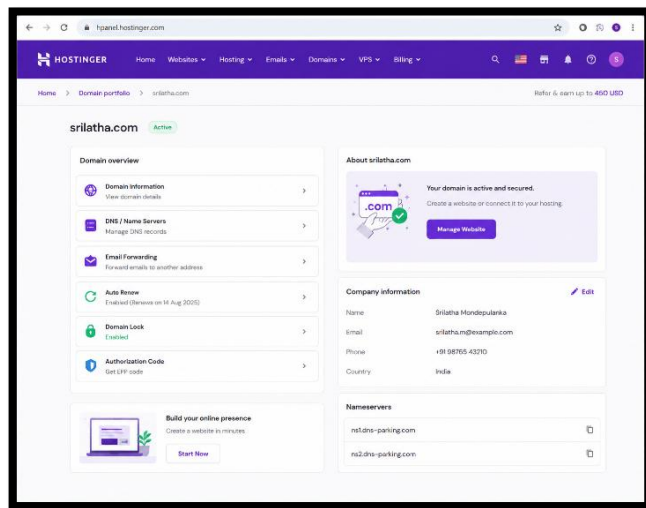
[+] No people found.

[+] Hosts found: 578
e.abx.com
e2f.abx.com
i.abx.com
imgserver.abx.com
imgserver.abx.com:192.288.117.42
i123.abx.com
i127.abx.com
i215.abx.com
i253.153.abx.com

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3. Technology Stack

- Web Server: Nginx
- Cloud Platform: AWS
- CDN: Amazon CloudFront
- SSL: AWS Certificate Manager



4. Public Subdomains

1. API
2. Help
3. Partner
4. Other public subdomains identified from Certificate Transparency logs.

crt.sh Identity Search							
Criteria Type: Identity Match: ILIKE Search: 'abc.com'							
Certificates	crt.sh ID	Logged At	Not Before	Not After	Common Name	Matching Identities	Issuer Name
	28558476970	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28558475441	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28558475196	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M01
	28558469094	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M01
	28558467608	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M01
	28558467788	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M01
	28558467107	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M01
	28558466573	2026-08-05	2026-08-05	2027-02-18	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28517253648	2026-08-04	2026-08-04	2027-02-17	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28514598663	2026-08-04	2026-08-04	2027-02-17	*.mydisney.abc.com	*.mydisney.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28489305799	2026-08-03	2026-08-20	2026-09-18	606166e16abc.com	*.606166e16abc.com 606166e16abc.com	C=US, O=Google Trust Services, CN=WE1
	284889564636	2026-08-03	2026-08-03	2027-02-16	api.partners.abc.com	api.abc.com api.partners.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28487842794	2026-08-03	2026-08-03	2027-02-16	api.partners.abc.com	api.abc.com api.partners.abc.com	C=US, O=Amazon, CN=Amazon RSA 2048 M04
	28417094804	2026-07-31	2026-07-31	2026-10-29	606166e16abc.com	*.606166e16abc.com 606166e16abc.com	C=US, O=SSL Corporation, CN=CloudFlare TLS Issuing ECC CA 4
	28273193162	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE2
	28273185138	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE2
	28273191651	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE2
	28273187365	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE2
	28272688513	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE2
	28272690134	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE2
	28272690144	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE1
	28272694753	2026-07-26	2026-07-26	2026-10-24	help.abc.com	help.abc.com	C=US, O=Let's Encrypt, CN=YE1

Topic 2: DNS Zone Transfers

Q2. (L3 – Apply) – 15 Marks

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.

Procedure :

- Identified the authoritative DNS servers using the `dig NS` command.
- Attempted a DNS Zone Transfer using the `dig AXFR` command.
- Analyzed the returned DNS records.

Command Used :

1. dig NS zonetransfer.me
2. dig AXFR zonetransfer.me @nsztm1.digi.ninja

Results

The DNS server allowed the AXFR request and returned 52 DNS records, including:

- Name Servers (NS)
- Mail Servers (MX)
- Internal Hosts
- Public Subdomains

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Topic 3: Introduction to Social Engineering – Social Engineering Attacks and Countermeasures

Q3. (L3 – Apply) – 20 Marks

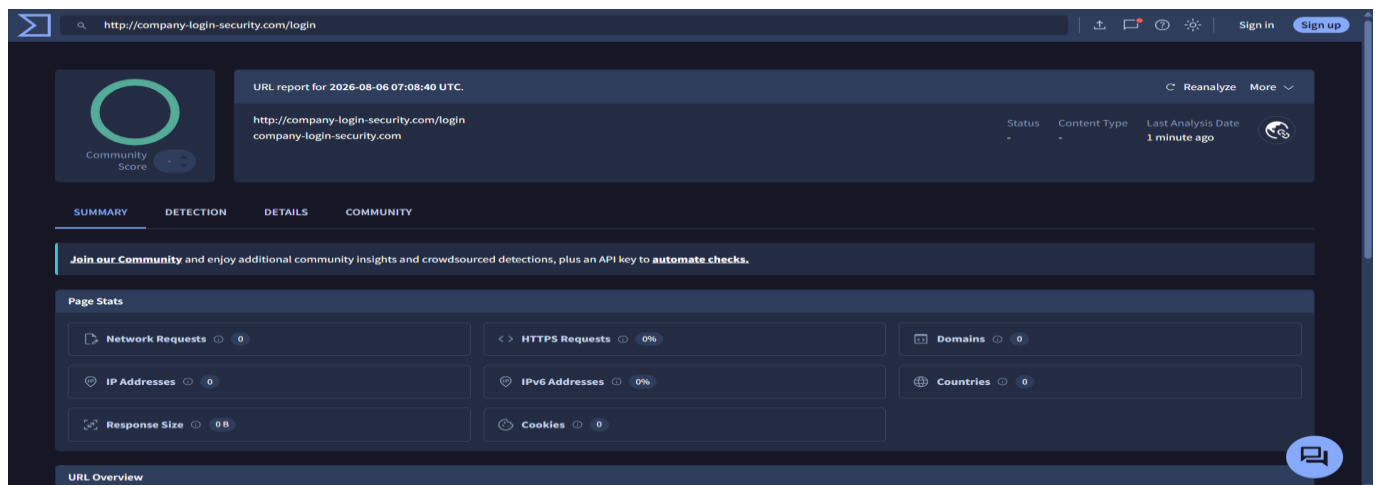
Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft

Phishing Attack :

Procedure :

- Examined the sender's email address.
- Verified the destination URL before clicking.
- Identified urgency and credential request.
- Recommended reporting the email to the IT department.



Countermeasures :

- Verify the sender's email address.
- Do not click suspicious links.
- Check the website URL before logging in.
- Confirm with the IT department.
- Enable Multi-Factor Authentication (MFA).
- Report suspicious emails immediately

Code of Conduct:

I M.Devi Srilatha (192525236) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Srilatha

RUBRICS FOR CALCULATION**Scenario Based Assessment**

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand